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Report of "Influence of surface tension implementation in Volume of Fluid and coupled Volume of Fluid with Level Set methods for bubble growth and detachment"

Type A project for NAPDE course

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1. Introduction

The study of bubble growth has a wide range of applications such as heat exchangers, stirring vessels, and flotation processes. In order to tackle the problem of bubble growth in viscous liquids, taking into account complex interface deformations and solid boundaries, moving and fixed grid methods have been developed. The article which I have been assigned to focuses on fixed grid methods; they assume that both fluids -namely the liquid and the gas- act as one mixture whose properties are determined from the interface position. In particular the authors developed two main interface capturing approaches: the Volume of Fluid (VOF) and Level Set (LS) methods. VOF and LS are then coupled in order to combine the advantages of the two methods.

1.1. Volume of Fluid (VOF) method

The volume of fluid (VOF) method is a free-surface modelling technique, i.e. a numerical technique for tracking and locating the free surface (the surface of a fluid that is subject to zero parallel shear stress, such as the interface between two homogeneous fluids). In VOF method the phase interface is represented by the fraction of liquid volume in each cell. The phase interface is moved by solving the following equation:

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \boldsymbol{v}) = 0 \tag{1}$$

where α is the liquid volume fraction, $\boldsymbol{v}$ is the velocity vector. It should be noted that because the standard advection schemes have too much dispersion, the artificial compression is usually used to preserve the jump in α. In VOF method the interface geometry can be reconstructed using the piecewise linear interface construction (PLIC) algorithm, which can be implemented following the mentioned steps: (1) calculate the normal direction to the interface $m = \nabla\alpha$; (2) assume the interface in each cell is planar; and (3) find a plane normal to m that has a liquid cell volume α.

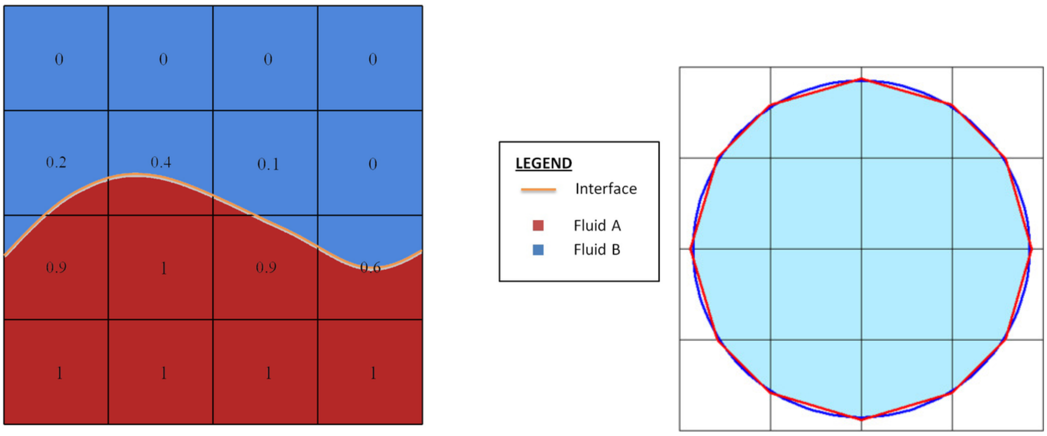
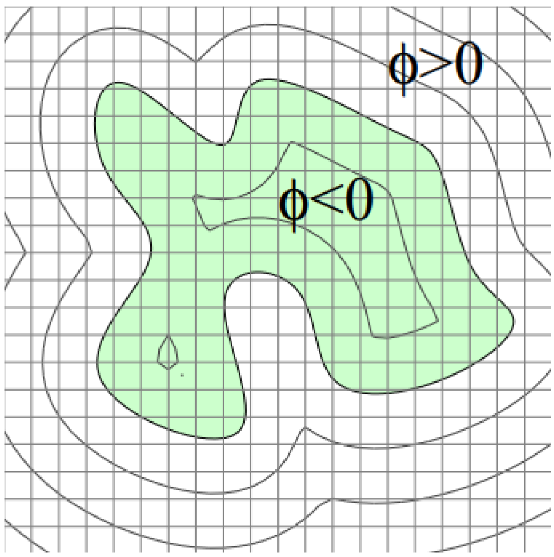


Figure 1: Function ϕ; reconstructed interface geometry by PLIC algorithm

It should be pointed out that the VOF method is not exactly volume preserving, in some extreme cases it is possible to find $\phi > 1$ or $\phi < 0$, or $\phi \neq 1$ in liquid or $\phi \neq 0$ in gas.

1.2. Level Set (LS) method

The level set method is a technique to represent moving interfaces or boundaries using a fixed mesh. It is useful for problems where the computational domain can be divided into two domains separated by an interface. The interface is represented by the 0-level set (or 0-isocontour) of a globally defined function, the level set function ϕ. ϕ is a smooth step function that is greater than zero in a domain and less than zero in the other. Across the interface, there is a smooth transition.



In two dimensions, the level-set method amounts to representing a closed curve Γ. Γ is represented as the zero-level set of ϕ by $\Gamma = \{(x, y)|\phi(x, y) = 0\}$. The level-set function satisfies the level set advection equation:

$$\frac{\partial \phi}{\partial t} + \boldsymbol{v} \cdot \nabla \phi = 0 \tag{2}$$

If the curve Γ moves in the normal direction with a speed v, then the level-set function ϕ satisfies the level-set Hamilton-Jacobi equation

$$\frac{\partial \phi}{\partial t} + v|\nabla \phi| = 0 \tag{3}$$

In fact the following calculations hold:

$$\Gamma = \{\boldsymbol{x} \,|\, \phi(\boldsymbol{x}) = 0\} \tag{4}$$

$$\phi(\boldsymbol{x}(t), t) = 0 \tag{5}$$

$$\frac{d}{dt}(\phi(\boldsymbol{x}(t),t)) = \nabla\phi(\boldsymbol{x}(t),t)\cdot\frac{d}{dt}(\boldsymbol{x}(t)) + \frac{\partial\phi}{\partial t}(\boldsymbol{x}(t),t) = 0 \tag{6}$$

$$\frac{\partial\phi}{\partial t}(\boldsymbol{x},t) + \boldsymbol{v}(\boldsymbol{x},t)\cdot\nabla\phi(\boldsymbol{x},t) = 0 \tag{7}$$

$$\boldsymbol{v} = v\cdot\frac{\nabla\phi}{|\nabla\phi|} \tag{8}$$

2. Numerical Formulation

2.1. Governing Equations

The governing equations for two isothermal, incompressible, immiscible fluids include the continuity, momentum, and interface capturing advection equations:

$$\nabla\cdot\boldsymbol{v} = 0 \tag{9}$$

$$\frac{\partial(\rho\boldsymbol{v})}{\partial t} + \nabla\cdot(\rho\boldsymbol{v}\boldsymbol{v}) = -\nabla P + \nabla\cdot\tau + \rho\boldsymbol{g} + \boldsymbol{F_\sigma} \tag{10}$$

$$\frac{\partial\alpha}{\partial t} + \nabla\cdot(\alpha\boldsymbol{v}) - \alpha(\nabla\cdot\boldsymbol{v}) = 0 \tag{11}$$

where ρ is the fluid density, $\boldsymbol{v}$ the fluid velocity vector, τ the viscous stress tensor defined as $\tau = 2\mu S = 2\mu(0.5(\nabla\boldsymbol{v}+\nabla\boldsymbol{v}^T)$, P the scalar pressure, $\boldsymbol{F_\sigma}$ the volumetric surface tension force, g the gravitational acceleration and α the interface capturing function. The fluid domain is defined for a single mixture where the function α is used to distinguish between the two fluids. The calculation of the volumetric surface tension force and the fluid physical properties, the density ρ and viscosity μ, vary according to the scalar field α: each quantity will be obtained performing a weighted average. Note that in (11) a compressive term is added to retain the conservativeness, convergence and boundedness.

2.2. Volume of Fluid

In the VOF method, α is the volume fraction function which takes value in $[0,1]$. The continuous fluid (liquid)is in the cells where $\alpha = 1$, while the dispersed fluid (gas) corresponds to $\alpha = 0$ The interface is smeared over the cells where $\alpha \in]0,1[$. The advection equation is re-formulated as:

$$\frac{\partial\alpha}{\partial t} + \nabla\cdot(\boldsymbol{v}\alpha) + \nabla\cdot(\boldsymbol{v_c}\alpha\beta) = 0 \tag{12}$$

where $\beta = 1-\alpha$, and $\boldsymbol{v_c} = \boldsymbol{v_l} - \boldsymbol{v_g}$ is the compressive velocity (where $\boldsymbol{v_l}$ is the liquid velocity and $\boldsymbol{v_g}$ is the gas velocity). Moreover, a compressive factor $c_\alpha$is used to increase compression as:

$$\boldsymbol{v_c} = min(c_\alpha|\boldsymbol{v}|, max|\boldsymbol{v}|)\frac{\nabla\alpha}{|\nabla\alpha|} \tag{13}$$

The authors used a constant compressive factor $c_\alpha = 2$. The physical properties of the liquid and the gas are computed using a weighted average as follows:

$$\rho = \rho_l\alpha + \rho_g(1-\alpha) \tag{14}$$

$$\mu = \mu_l\alpha + \mu_g(1-\alpha) \tag{15}$$

In the VOF method $\boldsymbol{F_\sigma}$ is defined as follows:

$$\boldsymbol{F_\sigma} = \sigma k(\alpha)\boldsymbol{\nabla}\alpha \tag{16}$$

where $k(\alpha)$ is the interface curvature calculated based on the updated value of α after advection:

$$k = -\boldsymbol{\nabla}\cdot(\hat{\boldsymbol{n}}_c\cdot\boldsymbol{S_f}) \tag{17}$$

where $\boldsymbol{S}_f$ is the surface vector of the cell face, f stands for the cell face, and $\hat{\boldsymbol{n}}_c$ is the unit interface normal which is calculated using the phase fraction field and it refers to the direction of the phase field changes in the numerical domain:

$$\hat{\boldsymbol{n}}_c = \frac{(\nabla\alpha)_f}{|(\nabla\alpha)_f|} \tag{18}$$

At wall boundary conditions, the phase fraction is determined so that a static contact angle θ_0 should be achieved at each time step, where this angle is defined as the angle between the interface normal and the face unit normal $\hat{\boldsymbol{n}}_{\boldsymbol{f}}$ at the wall:

$$\hat{\boldsymbol{n}}_c \cdot \hat{\boldsymbol{n}}_f = \cos\theta \tag{19}$$

The interface normal at the wall boundary condition is corrected to satisfy the static contact angle before calculating the curvature in (17).

2.3. Coupling VOF with LS (S-CLSVOF)

We would like to exploit the mass conservativeness of VOF and the interface smoothness of LS. In fact LS alone is not mass convervative and VOF alone does not allow to define a gradient and this could generate spurious currents at the interface. In the S-CLSVOF solver, a new Level Set field ϕ is introduced where the interface position is defined by the iso-line $\phi = 0$. Although two separate fields, namely VOF and LS, are defined to represent the fluid domain, only the VOF advection equation (12) is solved. The first step of the coupling is to assign an initial value to the Level Set function using the advected VOF fraction function and assuming that the interface position is defined at the iso-line contour $\alpha = 0.5$:

$$\phi_0 = (2\alpha - 1) \cdot \Gamma \tag{20}$$

where $\Gamma = 0.75\Delta x$ in order to have a value of ϕ close to the mesh step size. Note that ϕ_0 is a signed function: it has positive value in the liquid and a negative value in the gas. This value ϕ_0 is then re-distanced by solving the re-initialization equation (in general, in the context of LS method, re-distancing is essential to ensure that some properties are preserved, such as interface sharpness, numerical stability -without re-distancing, errors in the level set function can accumulate over time, leading to numerical instability-, and distance information):

$$\begin{cases} \frac{\partial \phi}{\partial \tau} = S(\phi_0)(1 - |\nabla\phi|) \\ \phi(\boldsymbol{x}, 0) = \phi_0(\boldsymbol{x}) \end{cases} \tag{21}$$

where τ is the artificial time step, $\boldsymbol{x}$ is the position vector, and $S(\phi_0)$ is a sign function. The solution converges to a signed distance function achieving $|\nabla\phi| = 1$. Since the Level Set function is a continuous function it helps in determining the interface normal as $\hat{\boldsymbol{n}} = \frac{\boldsymbol{\nabla}\phi}{|\boldsymbol{\nabla}\phi|}$. It then provides a smoother interface curvature. The volumetric surface tension can be calculated as $\boldsymbol{F_\sigma} = \sigma k(\phi)\delta(\phi)\boldsymbol{\nabla}\phi$ where σ is the surface tension coefficient and δ is defined as

$$\begin{cases} \delta(\phi) = 0 \text{ if } |\phi| > \epsilon \\ \delta(\phi) = \frac{1}{2\epsilon}(1 + cos(\frac{\pi\phi}{\epsilon})) \text{ if } |\phi| \le \epsilon \end{cases} \tag{22}$$

Defining $\delta(\phi)$ in this way means that $\boldsymbol{F_\sigma}$ will have an effect only around the interface: the influence of surface tension is spread over a finite interface thickness. Results have shown the parameter ϵ have a significant influence on the solution, however limiting the interface spread (i.e. taking a small value of ϵ) helps in achieving accurate models.

In the end the S-CLSVOF solver can be described in eight main steps:

1. Define vector and scalar fields for the multiphase flow problem.
2. Initialize the numerical fields, reinitialize the LS function and compute the initial values of the δ function.
3. Start the time loop by correcting the interface and the volume fraction.
4. Solve (12) and correct the new values of α at the boundaries where a static contact angle is imposed. Then, compute ϕ_0 using equation (20).
5. Re-initialize ϕ by means of (21). Then calculate the new values of the Dirac function.
6. Update the physical properties by means of equations (14) and (15)
7. Solve the Navier-Stokes equations for velocity and pressure.
8. Go back to step 3.

The governing equations are discretized based on a Finite Volume formulation. The mesh is a fixed uniform structured grid. The equations are integrated in time using the Euler implicit scheme, while the spatial derivatives are discretized using second order schemes.

3. Numerical Validation

S-CLSVOF is validated by studying a circular bubble at equilibrium in order to examine the spurious currents at the interface, and then by studying the free bubble rise (no interaction between the bubble free surface and the wall boundary condition takes place).

3.1. Circular Bubble at the Equilibrium

A circular bubble at equilibrium in a zero gravity field is used to assess the strength of the currents and to characterize the fluid flow in the absence of any external forces. The following values for physical properties are used: $\rho_g = 1\frac{kg}{m^3}$, $\mu_g = 10^{-5}\frac{kg}{ms}$, ρ_l=1000 $\frac{kg}{m^3}$, $\mu_l = 0.001\frac{kg}{ms}$, $\sigma = 0.01\frac{kg}{s^2}$. The mesh domain is 0.05×0.05m^2. The initial bubble radius is $R_{ini} = 0.005m$ is positioned at the center. Two different mesh resolutions are used: coarse and fine mesh. Coarse mesh has 10 cells per bubble diameter, fine mesh has 20. The time step size is 10^{-5} for the coarse mesh and $5\times10^{-6}s$ for the fine mesh. The results are considered at physical time 0.1 s. The error in the interface curvature and the norms of velocity are reported in the following table.

Table 1
Norms of velocity and curvature, and errors in the pressure jump at 0.1 s with time step $\Delta t = 10^{-5}$ s (Coarse mesh) and $\Delta t = 5 \times 10^{-6}$ s (Fine mesh).

Method	Mesh size (m)	l_{VOF}	$l_{S-CLSVOF}$	l_∞ (m/s)	l_1 (m/s)	E_0 (%)	E_{total} (%)	$E_{partial}$ (%)	E_{int} (%)
S-CLSVOF	0.001	–	0.2478	0.0171	0.00029	0.416	13.302	2.526	10.778
S-CLSVOF	0.0005	–	0.1527	0.0068	0.00011	0.875	7.866	2.198	5.668
VOF	0.001	1.1752	–	0.0129	0.00019	14.523	24.968	15.672	9.296
VOF	0.0005	1.3289	–	0.0261	0.00017	11.729	19.611	14.144	5.467

$$l_{VOF} = \sqrt{\frac{\sum_{i=1}^{M}(kR - k_{exact}R)^2}{M}} \tag{23}$$

$$l_{S-CLSVOF} = \sqrt{\frac{\sum_{i=1}^{N}(kR - k_{exact}R)^2}{N}} \tag{24}$$

where M is the number of cell in the numerical domain and N is the number of cells where the surface tension is calculated in the S-CLSVOF method, i.e. where $\delta \neq 0$.

$$l_\infty = \max_{i,j}(\|\boldsymbol{v}_{i,j}\|) \tag{25}$$

$$l_1 = \frac{1}{N_x N_y}\sum_{i,j}\|\boldsymbol{v}_{i,j}\| \tag{26}$$

These results show that the curvature error does not converge with grid refinement with either VOF or S-CLSVOF. However the error from the S-CLSVOF is one order of magnitude smaller than the VOF estimate. This is most likely due to the reduction in the interface thickness and the resulting focussing of surface tension towards the central part of the interface. Refining the mesh and taking $\epsilon = 1.5\Delta x$ also increases the maximum value of δ, as defined by (22), at the center of the interface region. This can explain the reduction in spurious currents as shown in the following figure. The plots show a more gradual variation in the velocity field within the bubble with significantly lower velocity magnitudes predicted by S-CLSVOF while large velocities localized over specific areas of the bubble with the VOF method.

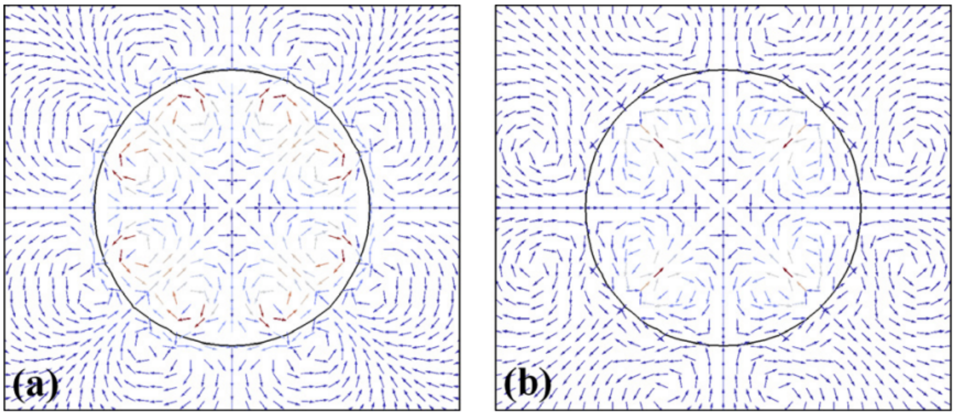


Fig. 2. Velocity vectors plot at 10th time steps, $\Delta t = 1 \times 10^{-05}$ s and $\Delta x = 5 \times 10^{-04}$ m for (a) S-CLSVOF (max. velocity 0.0068 m/s) and (b) VOF (max. velocity 0.026 m/s). The color map varies from dark blue (min velocity) to dark red (max velocity).

3.2. Free Bubble Rise

Three different dimensionless numbers influence the shape of the bubble: the Morton number ($Mo = g\mu_l^4 \Delta\rho/\rho_l^2\sigma^3$) the Eötvös number ($Eo = g\Delta\rho D_{eq}^2/\sigma$), and the Reynolds number ($Re = \rho_l V_\infty D_{eq}/\mu_l$). The authors of the article validated the S-CLSVOF solver by modeling a single air bubble rising in a viscous liquid. The bubble does not interact with the boundary conditions and the accuracy of the numerical methods can be assessed by comparing against broadly available experimental data. The single bubble rise is simulated using the VOF and S-CLSVOF methods. Results with respect to different physical properties are reported in the following tables and plots. A regular mesh is used to discretize the fluid domain so that 25 cells are distributed along the bubble diameter. This mesh size has been shown to be sufficient for guaranteeing the convergence of numerical results.

Table 3
Physical properties used for numerical bubble rise simulations.

Series	μ_l (Pa s)	ρ_l (kg/m^3)	σ (N/m)	Mo
S1	0.687	1250	0.063	7.5287
S3	0.242	1230	0.063	0.1057
S5	0.0733	1205	0.064	7.4492×10^{-4}

Table 4
Simulation parameters for the rising of different sized bubbles in the series fluids.

Diameter (m)	Bubble					
	S1		S3		S5	
	Mo	Eo	Mo	Eo	Mo	Eo
0.003	7.529	1.75	0.106	1.7221	7.4492×10^{-4}	1.6607
0.005	7.529	4.862	0.106	4.7836	7.4492×10^{-4}	4.6131
0.007	7.529	9.529	0.106	9.3759	7.4492×10^{-4}	9.0416

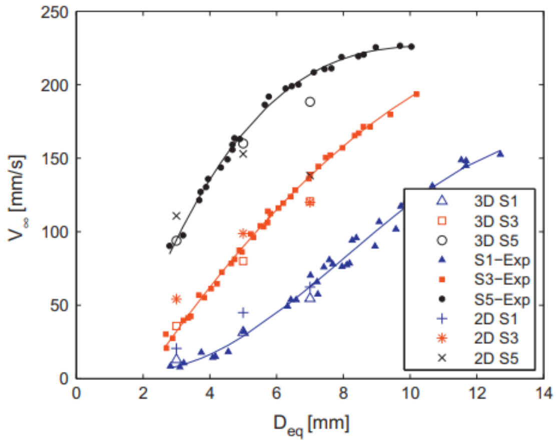


Fig. 4a. Comparison of bubble terminal velocity predicted by simulations (2D and 3D S-CLSVOF) with experimental observations (Raymond and Rosant, 2000).

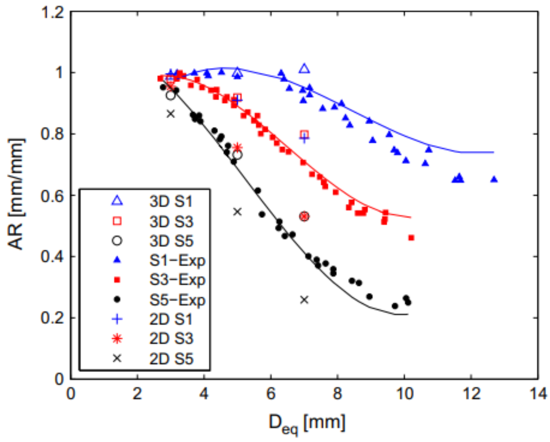


Fig. 4b. Comparison of bubble aspect ratio predicted by simulations (2D and 3D S-CLSVOF) with experimental observations (Raymond and Rosant, 2000).

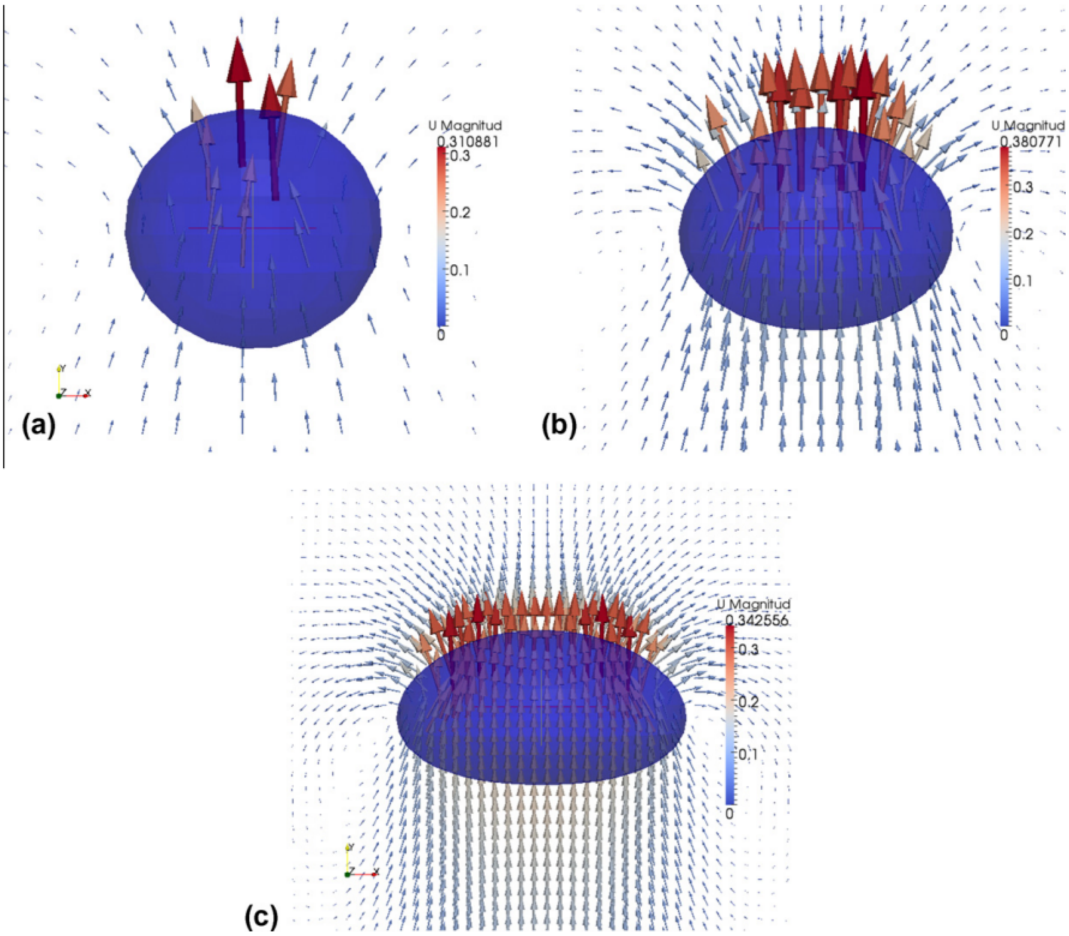


Fig. 3. The bubble shape and the velocity vector plots for three diameters (a) D_{eq} = 0.003 m, (b) D_{eq} = 0.005 m, (c) D_{eq} = 0.007 m predicted by S-CLSVOF at time 0.14 s.

Bubble terminal velocity increases as larger equivalent diameters are considered. For small bubble diameters, the 2D simulation gives reasonable bubble terminal velocities, however it fails to predict the correct V_∞ at high equivalent diameters, contrary to 3D simulations. This highlight the importance to model the free large bubble rise with 3D simulations. Bubble rise is in fact 3D in nature as the wake drives the bubble to follow a 3-directional trajectory during the rise. The following table reports a comparison between the VOF and S-CLSVOF method. No significant differences are reported as both methods well predict the terminal velocity and aspect ratio.

Table 5
Relative difference between the bubble terminal velocity and aspect ratio from the S-CLSVOF method compared to the VOF predictions.

Diameter (m)	Bubble					
	S1		S3		S5	
	E_V (%)	E_{AR} (%)	E_V (%)	E_{AR} (%)	E_V (%)	E_{AR} (%)
0.003	2.604578	0.406504	−0.05574	1.661475	−1.93252	1.982379
0.005	1.744186	1.422764	2.564103	2.33853	2.210015	1.949861

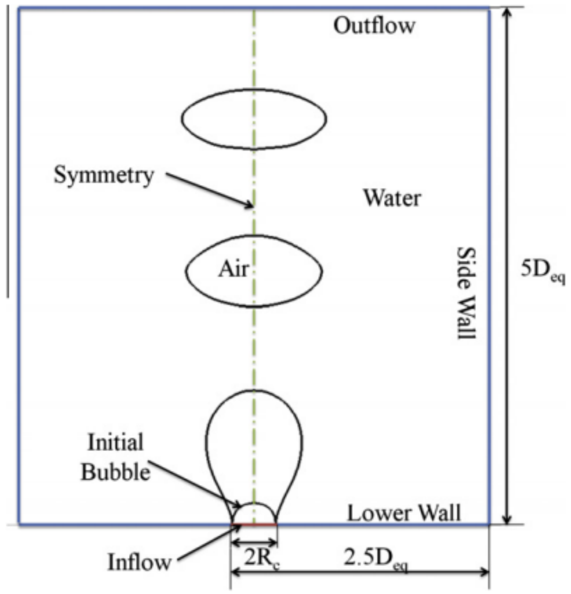


Fig. 5. Schematic diagram of the experimental rig and computational boundary conditions.

4. Bubble Growth and Detachment

4.1. Computational setup

The bubble formation is studied axi-symmetrically since it grows vertically without any lateral oscillations. As shown by the figure above, the air bubble is injected through an orifice of radius $R_c = 0.8 \times 10^{-3} m$, the surface tension coefficient and the physical properties of both liquid and gas are supposed to be constant. The gas injection is supposed to be under constant flow rate $\dot{Q} = 2.778 \times 10^{-8} m^3/s$. The bubble passes through two stages during its formation: the expansion stage and the collapse (detachment). The bubble detachment happens when the neck diameter is less than 10% of the orifice diameter. The following table shows the performed convergence analysis of mesh discretization. The results show that the bubble detachment characteristics increases with mesh refinements. Note that $\Delta t = 0.2 \Delta x$.

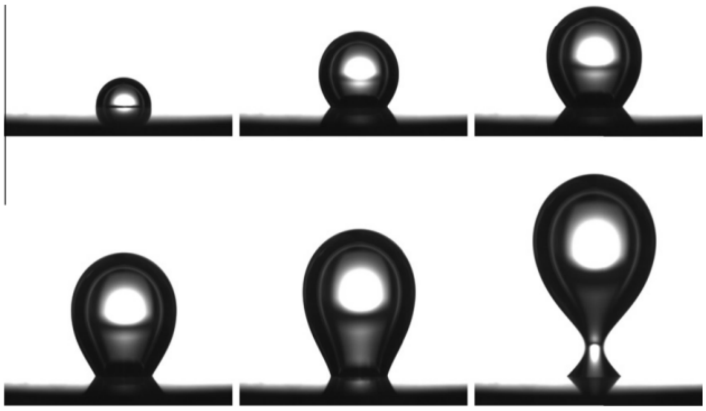
Convergence analysis of mesh discretization using S-CLSVOF with R_c = 0.8 mm and $\dot{Q} = 200$ mlph.

Nb/ diameter	Δx	Δt	V_{det} (mm^3)	t_{det} (s)	E_{vol} (%)	E_t (%)
8	2.0×10^{-4}	4.0×10^{-5}	26.595	0.455	–	–
16	1.0×10^{-4}	2.0×10^{-5}	28.082	0.484	5.30	6.01
20	8.0×10^{-5}	1.6×10^{-5}	30.283	0.525	7.27	7.79
32	5.0×10^{-5}	1.0×10^{-5}	31.448	0.547	3.70	4.02
64	2.5×10^{-5}	5.0×10^{-6}	31.985	0.560	1.68	2.32

Four boundary conditions are set to represent the borders of the numerical domain. The inflow velocity is defined at the inlet where the gas is injected. A parabolic profile is set for the inflow velocity. At the wall a no slip condition is imposed except for the lower wall where wall adhesion is considered. The wall static angle is $\theta = 20°$ so that the bubble interface does not spread along the wall. At the initial time, a semi-circular bubble is patched at the inflow with a radius $R_i = R_c = 0.8 \times 10^{-3} m$. The existence of the initial bubble is essential to compute the LS function. S-CLSVOF method requires ϵ as an additional parameter to determine the thickness of the influence of surface tension. The condition $\epsilon \geq 1.5\Delta x$ guarantees that surface tension is spread over at least one cell on either side of the interface.

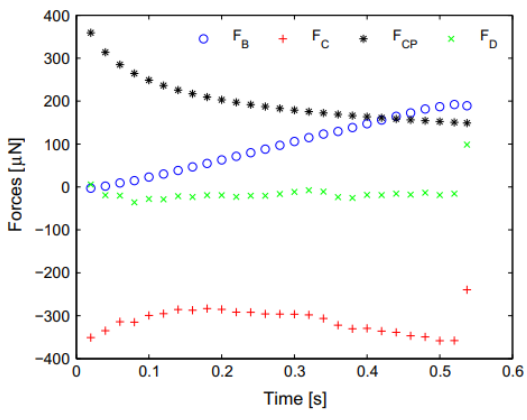
4.2. Experimental bubble formation

The numerical results obtained by VOF and S-CLSVOF are first assessed qualitatively by comparing the contour of the reconstructed interface against experimental observations. The image below shows the snapshot at six different time frames. Note that during the detachment stage, three different regions can be distinguished: the bubble dome which forms the upper part, the neck region which forms the lower part, and the transition region where the bubble interface curvature converts from concave to convex.



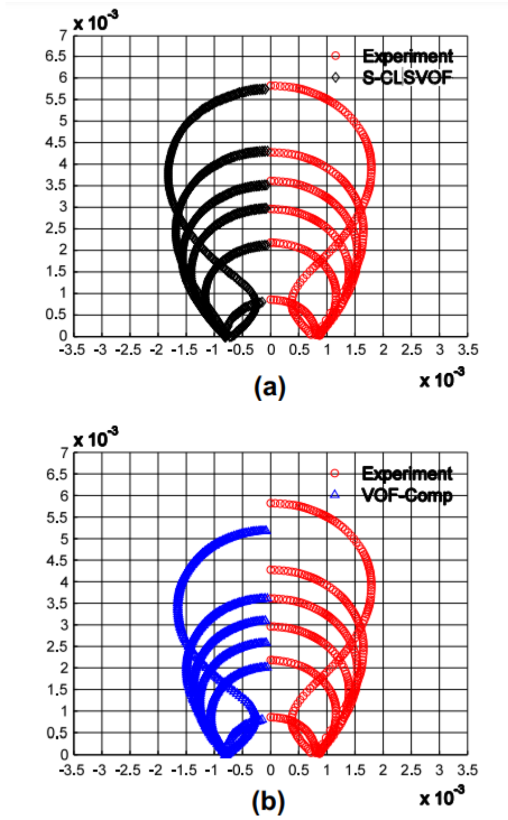
Experimental bubble shape at six different frames $t/t_{det} \sim 0, 0.2, 0.4, 0.6, 0.8, 1$ ordered from top left to bottom right with $R_c = 0.8$ mm, $\dot{Q} = 200$ mlph.

Under quasi-static conditions, the forces acting on the bubble are divided into two main groups: the first group promotes the bubble detachment and includes the buoyancy force and the contact pressure force, the other group of forces resists bubble detachment and acts to keep the bubble attached to the orifice. It is a combination of the capillary force and the dynamic force which respresents the liquid inertia close to the bubble interface and the viscous forces. During the growth, the influence of the buoyancy increases while at the last stage of detachment, the influence of the capillary force decreases rapidly. The forces acting on the bubble and their variation in time are represented in the plot below. The influence of the capillary force highlights the importance of using accurate surface tension model in order to avoid unphysical solutions.



Forces acting on the bubble during the formation process including: the buoyancy force (F_B), the capillary force (F_c), the contact pressure force (F_{CP}), and the dynamic force (F_D) with $R_c = 0.8$ mm and $\dot{Q} = 200$ mlph.

A comparison between the experimental bubble and the numerical interfaces predicted by both VOF and S-CLSVOF is presented in the images below. With VOF the bubble is shown to grow faster compared to the experimental observations. In contrast S-CLSVOF is in close agreement with the experimental measurements. This is probably due the ϵ-implementation of surface tension: in fact it was shown that increasing ϵ (i.e. increasing the interface smearing) tends to provide detachment parameters which converge towards those predicted by VOF.



5. Take-home message

After a careful study of this article I would like to highlight those aspects that I consider key in the implementation of S-CLSVOF method: the level set method provides a smoother interface in contrast to VOF, this is key in order to define the δ function so that we can limit the influence of surface tension just to the interface and not on the whole domain. This improves the agreement with experimental observation, in fact the capillary force is the predominant force during the bubble growth. However, LS method alone would not be enough. LS is in fact used just to obtain the smoother interface but only the VOF equation is solved. This is due to the fact that LS alone does not preserve mass conservation (it should be pointed out that LS is mass conservative at the limit of mesh refinement). Moreover VOF method used with PLIC (piecewise-linear interface calculation) is a contemporary standard and it is what is implemented in OpenFoam (the open source software used by the authors). On the other hand, the main drawback of VOF is that it tends to smear the free surface due to the excessive diffusion introduced by the transport equation and to the fact that the advection equation is solved for a discontinuous function. Another aspect that can be critical in VOF method is the evaluation of the interface curvature, essential in applications where surface tension effects are relevant. These problems, as previously said, are tackled coupling VOF with LS (which better tracks the interface between gas and liquid).

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